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CASE STUDY · CITY OF OCEANSIDE, CALIFORNIA

The Problem MARCIA Was Built to Solve

Case study: how a California municipality achieved one-year surveillance retention on a city budget using multi-tier disk-and-LTO storage



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At a glance

Customer: City of Oceanside, California, San Luis Rey Wastewater Treatment Plant

Requirement: keep every frame of video for **one year**, as California records rules require for government records

Scale: more than **30 high-definition (1080p) IP cameras** recording **24/7**, about **548 TB** of video for one year

Result: one-year retention met, within the City's budget, at **54% lower five-year cost** than a disk-only system, with operators playing back any recording themselves, without IT

About this case study. The Oceanside deployment used Cozaint's askALICE™ system: a disk tier for recent video and an LTO tape tier for long-term retention, managed by the video management software. It proved, in daily use at a working city facility, the multi-tier disk-and-LTO architecture that Cozaint's patented MARCIA™ Intelligent Video Lifecycle Management platform is built on. Oceanside is a customer of askALICE; this document does not claim it runs MARCIA.

The challenge

Keeping a city's water treatment plant safe and secure matters to many people: residents, regulators, City staff and the operators who run the plant. The San Luis Rey Wastewater Treatment Plant records its cameras around the clock, and the City connected another 14 cameras from a second location, for more than 30 camera feeds in all.

California's state records management rules require certain government records to be kept for one year, and video surveillance files can be government records. For the City, long-term retention was therefore both a regulatory duty and a security need.

Sizing one year of continuous 1080p video from more than 30 cameras came to roughly **548 terabytes**. A traditional hard-drive-only system of that size was simply too expensive for the City's budget.

> "We needed to upgrade our security surveillance system at our water treatment plant, and since we have worked with several different systems, we knew what we liked and what we didn't like. It is always difficult to find a system that fits within our budget and meets our long-term retention requirements. Cozaint's askALICE has been an ideal fit for our surveillance recording needs and our budget."
>
> Scott Prather, Senior IT Analyst, City of Oceanside, California

Why the usual work-arounds were not enough

The surveillance industry has tried many ways to squeeze video onto less disk:

- **Better compression (H.264, H.265, H.265+).** It helps, but as camera counts and resolutions grow, the storage need keeps growing and the cycle starts again.
- **Record-on-motion.** Trees moving, birds and flickering lights trigger recording, so vendors added thresholds. The result: systems that sometimes did **not** record when something important happened.
- **Lower resolution, frame rate or retention.** All of these save disk by throwing away the evidence the cameras were bought to capture.

Every one of these work-arounds treats the video as the problem. The City needed the opposite: keep full-quality video for the full year, and make the storage affordable.

The approach: put each part of the video lifecycle on the right storage

The solution was to stop keeping everything on spinning disk. The deployment used two tiers:

Tier	What it holds	Why
Tier 1: disk	The most recent video	Instant playback for the incidents operators review most often
Tier 2: LTO tape	Every recording, for the full retention period	Very low cost per terabyte, decades of archive life, no power when a cartridge is at rest

New video was replicated to LTO automatically in the background. Video for each camera was kept together on the same cartridges, which makes recalling a given camera's history efficient.

The part that made it work: the operator never leaves the VMS

Tape has been used to archive video for decades in Media & Entertainment. It had not caught on in surveillance for one reason: operators do not want to leave their video management software (VMS), find a tape and wait for an IT restore job.

At Oceanside, operators simply moved the VMS timeline to the date and time they needed. If that video was on disk, it played at once. If it was on LTO, the system located the right cartridge, the tape library loaded it, and the video came back to the same timeline, typically within a few minutes, with no extra steps and no IT staff involved.

From Oceanside to MARCIA. That timeline-driven recall from LTO is the workflow Cozaint went on to patent. MARCIA™ (U.S. Patent No. 12,755,902) brings it to the VMS platforms enterprises already run, including Milestone XProtect, Hanwha WAVE, Digital Watchdog DW Spectrum, Network Optix Nx Witness and Cozaint BOBBYvms.

The results

One year of full-quality video, within budget

The City kept a full year of 1080p video from every camera without cutting resolution, frame rate or retention, and within its budget allotment.

54% lower five-year cost than disk-only

Costs were calculated with the FUJIFILM video surveillance retention cost calculator, using the plant's own camera count, resolution and retention needs.

Five-year total cost of ownership	Two-tier (disk + LTO)	Disk-only
Tape drives, library and media	$20,618	$0
Disk storage	$7,318	$59,993
Energy	$3,579	$8,513
Cloud service fees and bandwidth	$0	$0
Five-year total	**$31,515**	**$68,506**

The two-tier design cost **less than half** as much as the disk-only design: a saving of 54%. Most of the difference comes from replacing a large disk array with a small disk tier plus LTO, and from the energy a cartridge on a shelf does not use.

The security team works on its own

Operators can play back any recording from any point in the year themselves. The IT team, which does not run the security operation day to day, no longer has to be called in to find or restore video. That saves the City operating cost as well as time during investigations.

> "The askALICE solution has provided the features and functionality our team needs from a video surveillance solution together with the long-term retention needs required by our City policies."
>
> Scott Prather, Senior IT Analyst, City of Oceanside, California

Proven in daily operation

The system has run in production at the San Luis Rey Wastewater Treatment Plant, meeting the City's records retention policy while giving the security and IT teams quick access to every recording.

What Oceanside means for your organization

Oceanside's situation is common: a mandated or desired retention period, more cameras at higher resolution, and a budget that disk-only storage cannot meet. The lessons carry over to banks, casinos, airports, hospitals, prisons, campuses and municipalities:

1. **Do not shrink the evidence to fit the storage.** Keep full resolution and frame rate, and change the storage instead.
2. **Size a small disk tier for recent video** and put long-term retention on LTO.
3. **Insist on timeline playback.** If operators must leave the VMS or call IT to see older video, the archive will not be used.
4. **Compare five-year cost, not just purchase price.** Energy, cooling and disk replacement change the answer.

Today, MARCIA delivers this architecture as a patented platform that works with the VMS you already run.

See your own numbers. Run the storage cost calculator at **cozaint.com/assessment**, or contact Cozaint at **info@cozaint.com** or **760-975-8000**.

The original 2023 askALICE case study remains available from Cozaint on request. Product names are trademarks of their respective owners. MARCIA is protected by U.S. Patent No. 12,755,902.